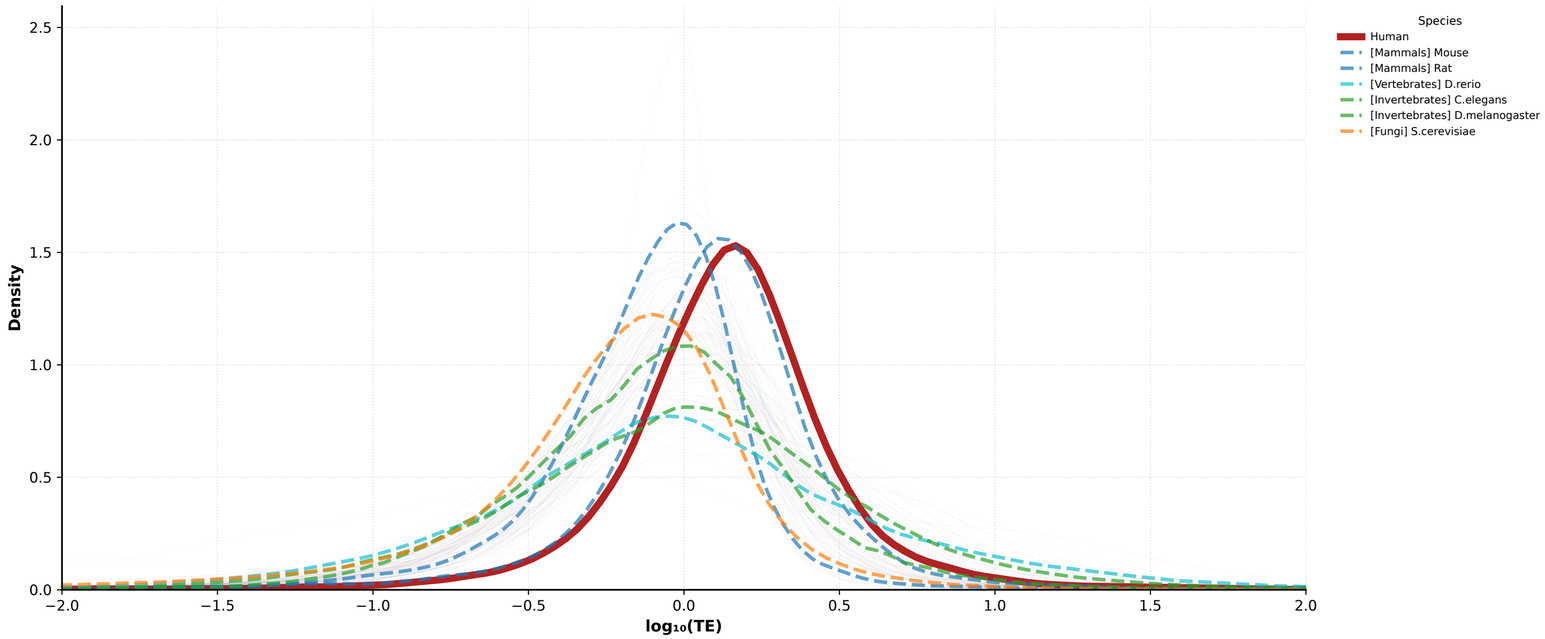
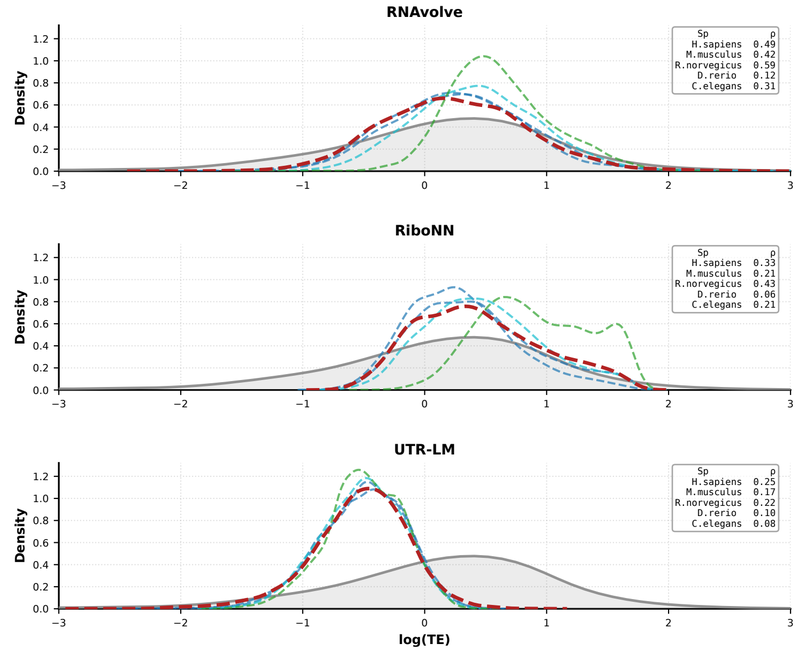
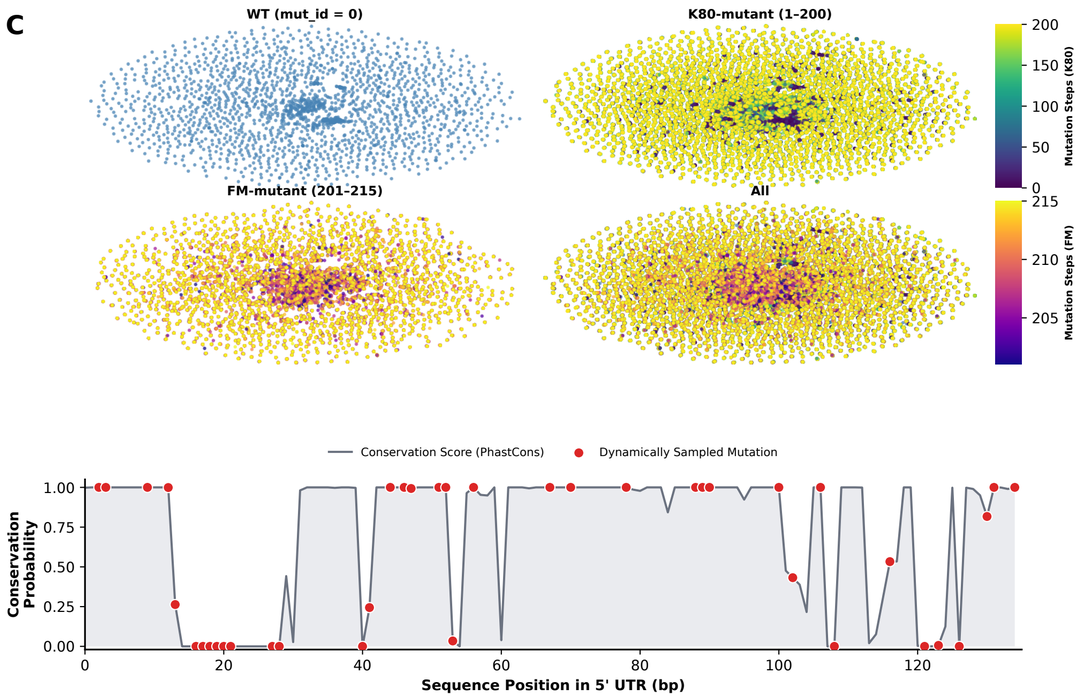
A


Species
Human
[Mammals] Mouse
[Mammals] Rat
[Vertebrates] D.rerio
[Invertebrates] C.elegans
[Invertebrates] D.melanogaster
[Fungi] S.cerevisiae
Density
log₁₀(TE)

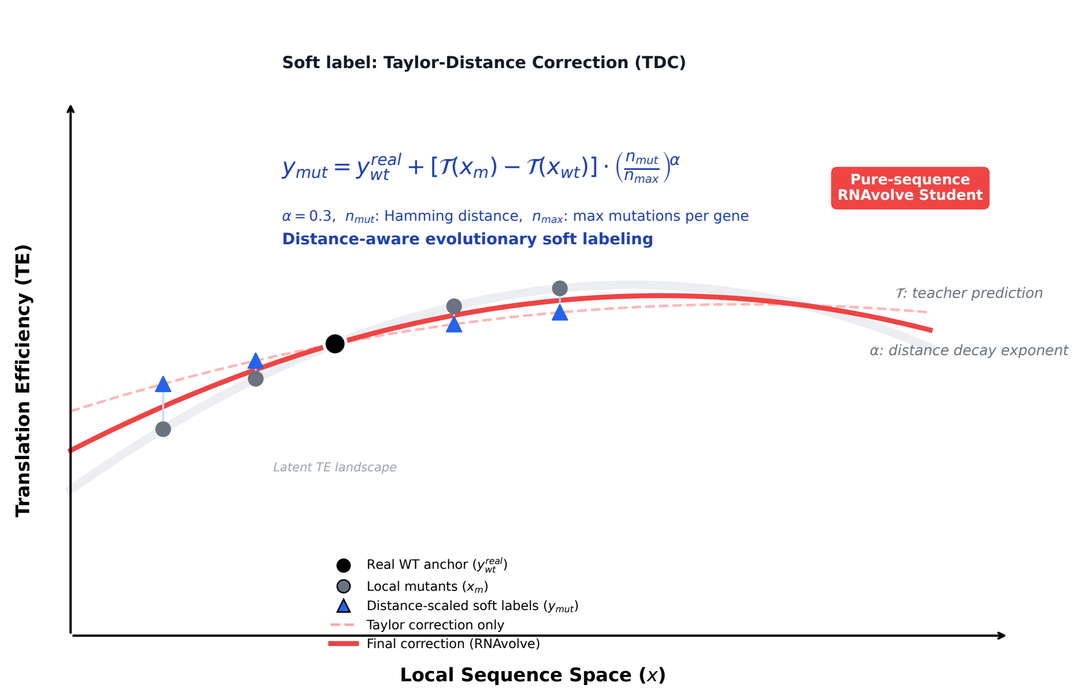
B


RNAvolve
Sp ρ
H.sapiens 0.49
M.musculus 0.42
R.norvegicus 0.59
D.rerio 0.12
C.elegans 0.31
RiboNN
Sp ρ
H.sapiens 0.33
M.musculus 0.21
R.norvegicus 0.43
D.rerio 0.06
C.elegans 0.21
UTR-LM
Sp ρ
H.sapiens 0.25
M.musculus 0.17
R.norvegicus 0.22
D.rerio 0.10
C.elegans 0.08
Density
log(TE)

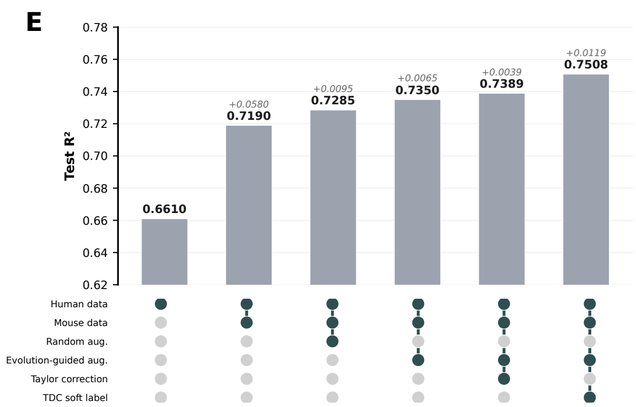
C


WT (mut_id = 0)
K80-mutant (1–200)
FM-mutant (201–215)
All
Mutation Steps (K80)
Mutation Steps (FM)
Conservation Score (PhastCons)
Dynamically Sampled Mutation
Conservation Probability
Sequence Position in 5' UTR (bp)

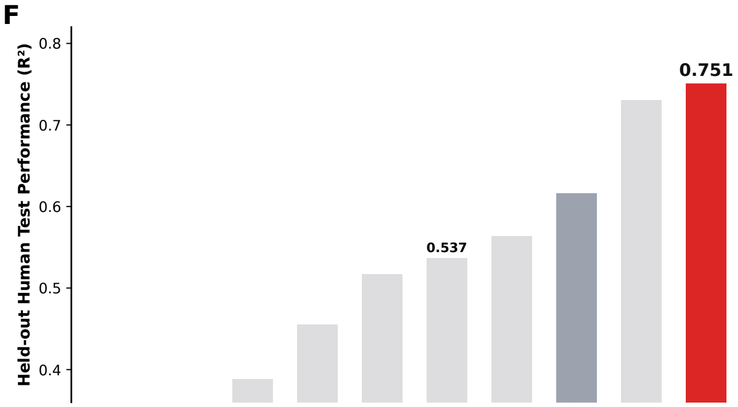
D


Soft label: Taylor-Distance Correction (TDC)
$y_{mut} = y_{wt}^{real} + [\mathcal{T}(x_m) - \mathcal{T}(x_{wt})] \cdot \left(\frac{n_{mut}}{n_{max}}\right)^{\alpha}$
α = 0.3, $n_{mut}$: Hamming distance, $n_{max}$: max mutations per gene
Distance-aware evolutionary soft labeling
Pure-sequence RNAvolve Student
Translation Efficiency (TE)
Local Sequence Space (x)
T: teacher prediction
α: distance decay exponent
Latent TE landscape
Real WT anchor
Local mutants
Distance-scaled soft labels
Taylor correction only
Final correction (RNAvolve)

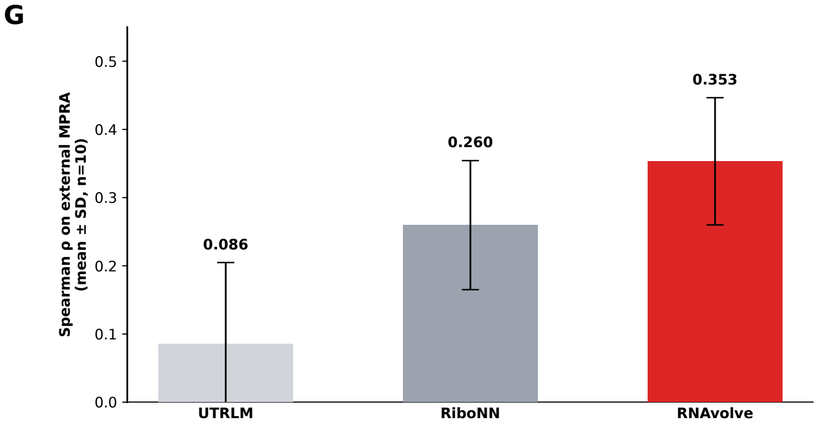
E


Test R²
0.6610
+0.0580
0.7190
+0.0095
0.7285
+0.0065
0.7350
+0.0039
0.7389
+0.0119
0.7508
Human data
Mouse data
Random aug.
Evolution-guided aug.
Taylor correction
TDC soft label

F


Held-out Human Test Performance (R²)
0.537
0.751
UTR-LM
mRNABERT
RNA-FM
LucaOne
CodonBERT
DNABERT-2
Orthrus
RiboNN
Evo2
RNAvolve

G


Spearman ρ on external MPRA
(mean ± SD, n=10)
0.086
0.260
0.353
UTRLM
RiboNN
RNAvolve